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(54) SYSTEMS AND METHODS FOR A BRA FOR USE WITH LIMITED DEXTERITY

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A41F 1/00 (2006.01)
A41C 3/02 (2006.01)
A44B 11/25 (2006.01)

(52) **U.S. Cl.**

CPC ***A41F 1/006*** (2013.01); ***A41C 3/02*** (2013.01); ***A41F 1/002*** (2013.01); *A44B 11/25* (2013.01)

(58) **Field of Classification Search**

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USPC 450/58
See application file for complete search history.

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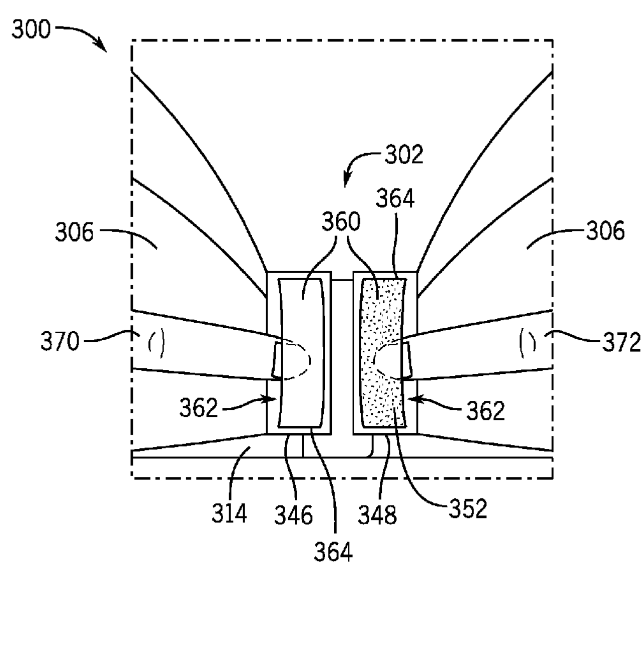
(57) **ABSTRACT**

A garment, such as a bra, can include a closure system that facilitates operation by a user with limited dexterity. The closure system can include a first clasp member, a second clasp member, a first finger pocket dimensioned to receive a first finger, and a second finger pocket dimensioned to receive a second finger. In use, the wearer can insert first and second fingers into the respective finger pockets and pinch or otherwise move their fingers toward each other to move the first and second clasp members together. The first and second clasp members can be selectively coupled or decoupled to secure or remove the garment from the wearer.

17 Claims, 11 Drawing Sheets



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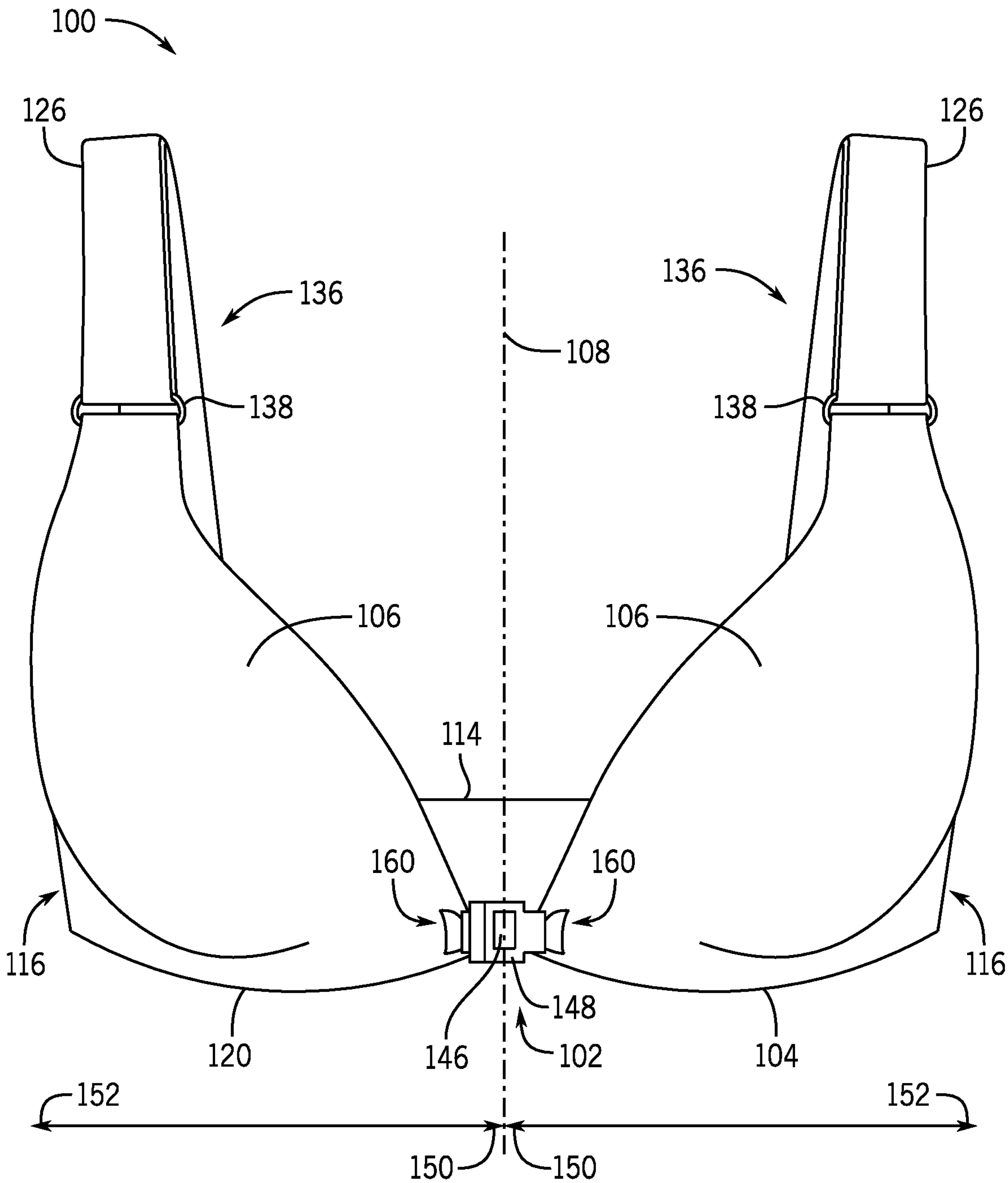
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FIG. 1

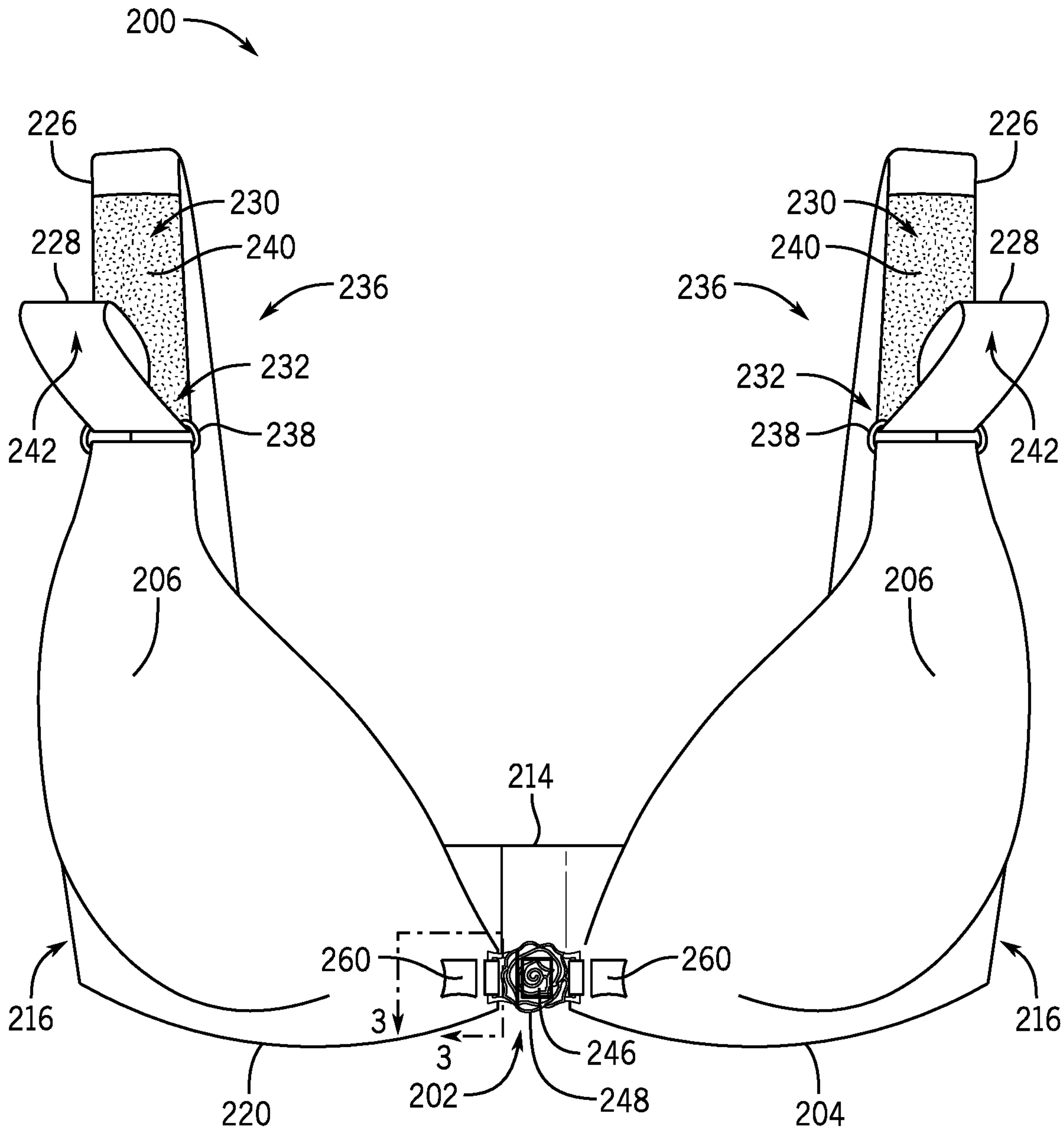
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FIG. 2

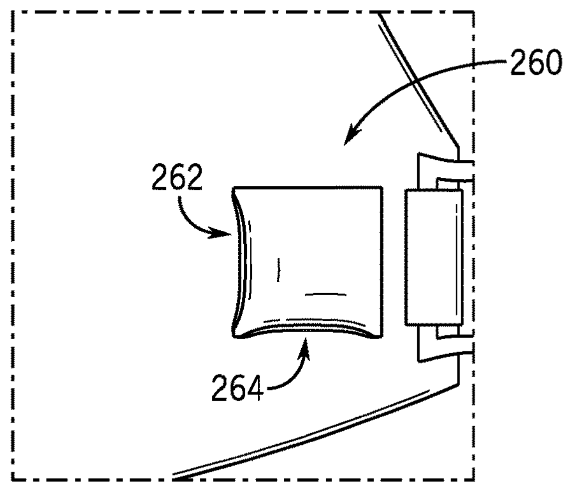
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FIG. 3

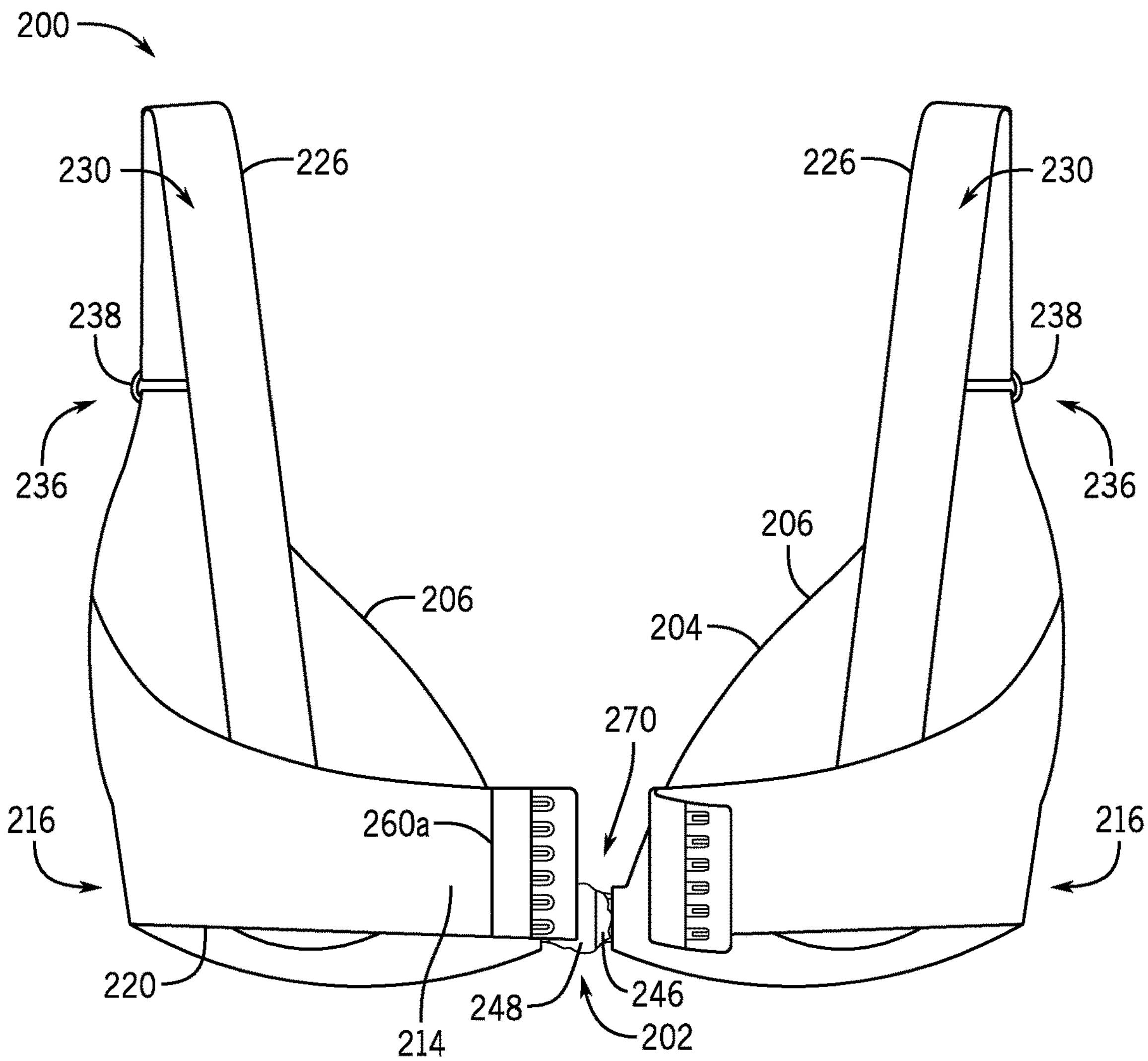
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FIG. 4

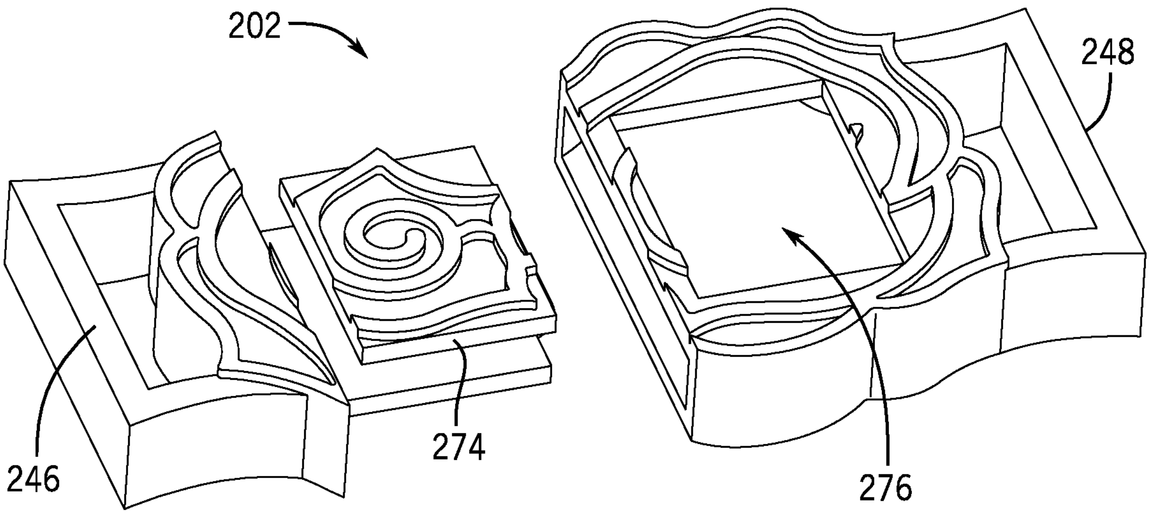

FIG. 5A

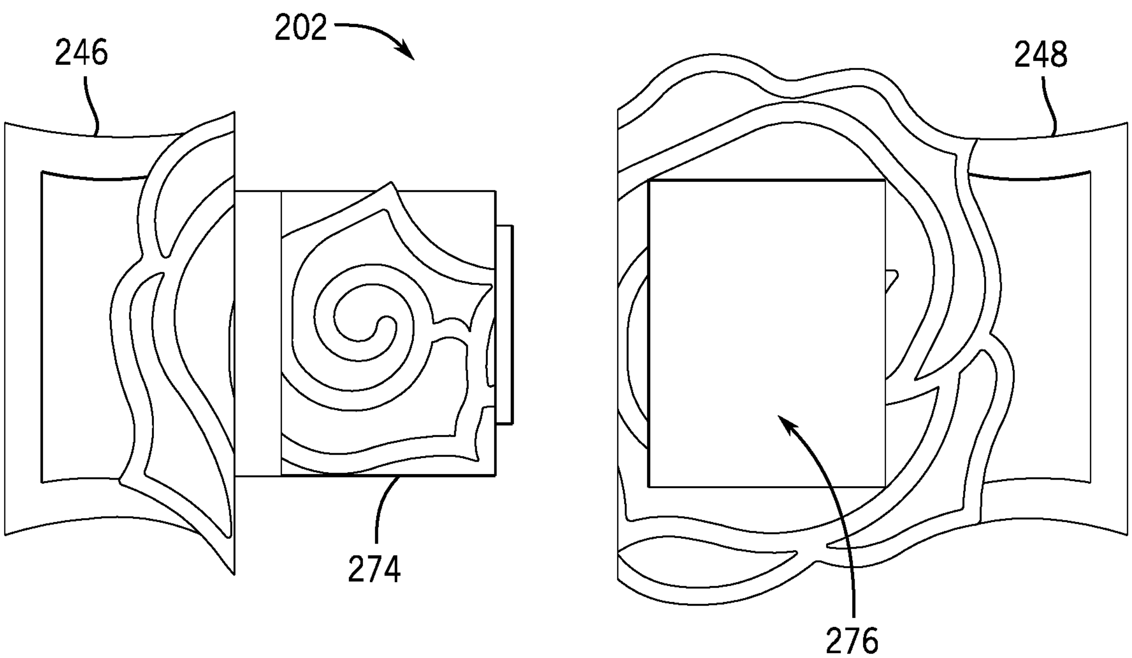

FIG. 5B

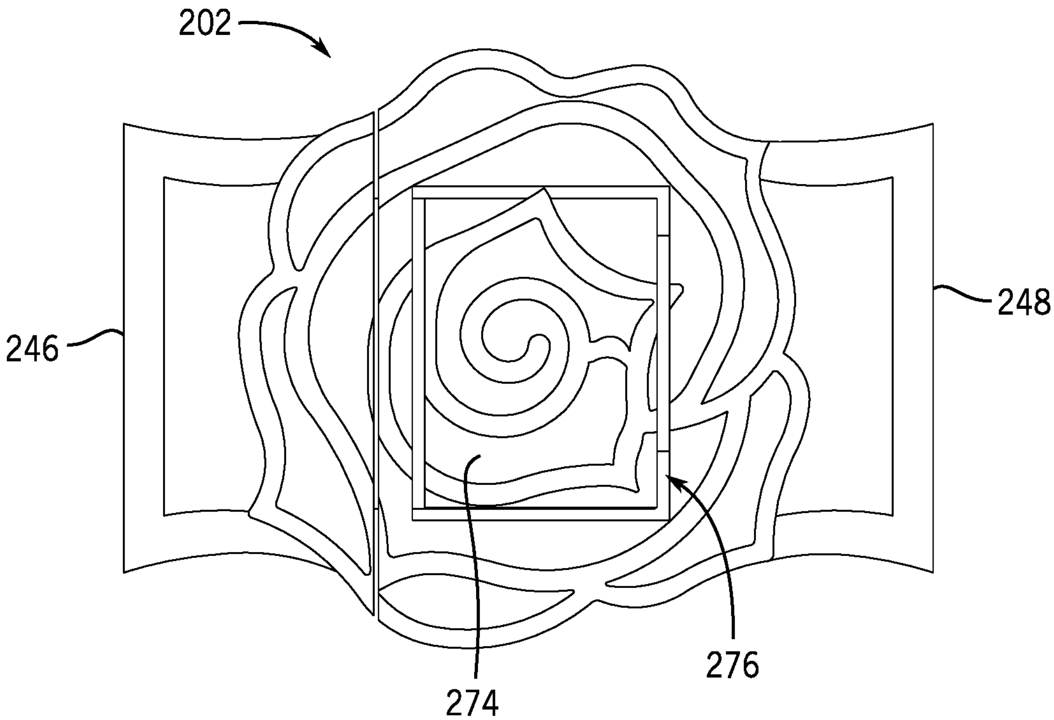

FIG. 5C

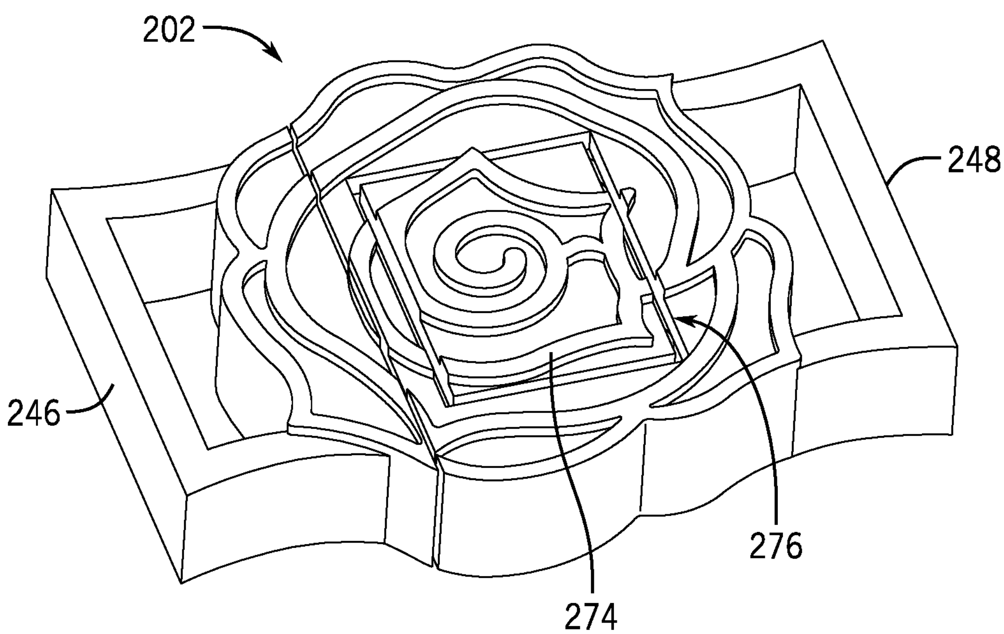

FIG. 5D

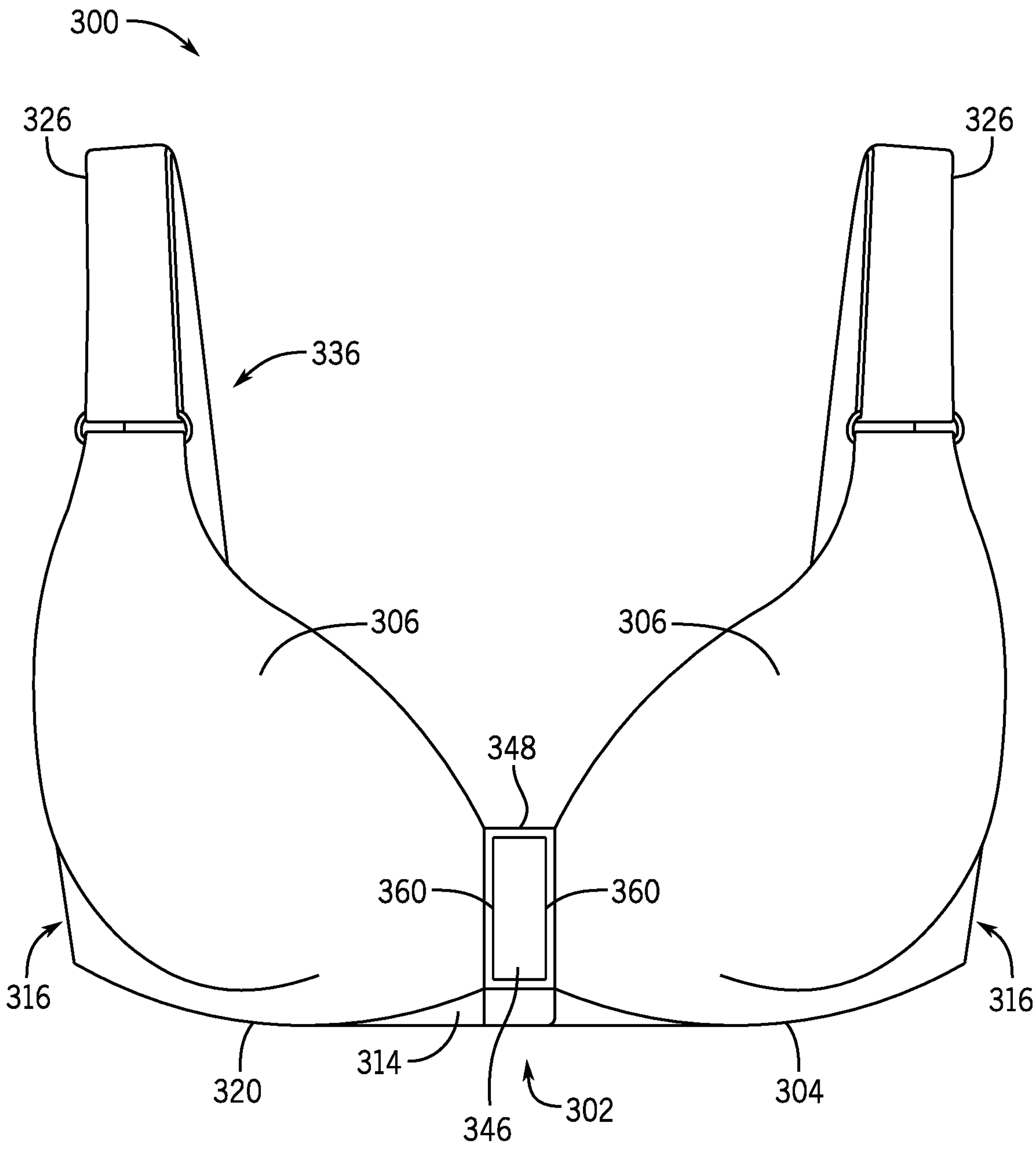

FIG. 6

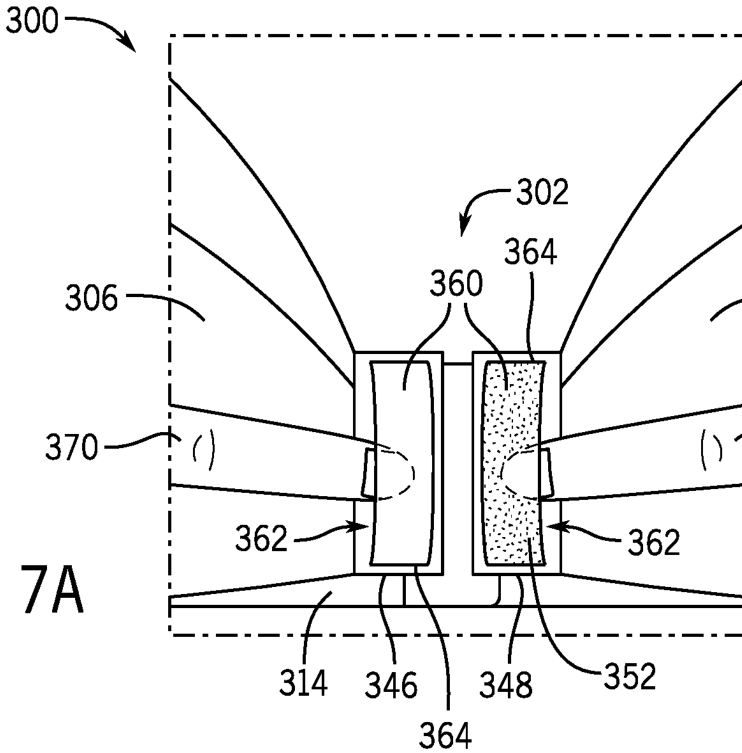
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FIG. 7A

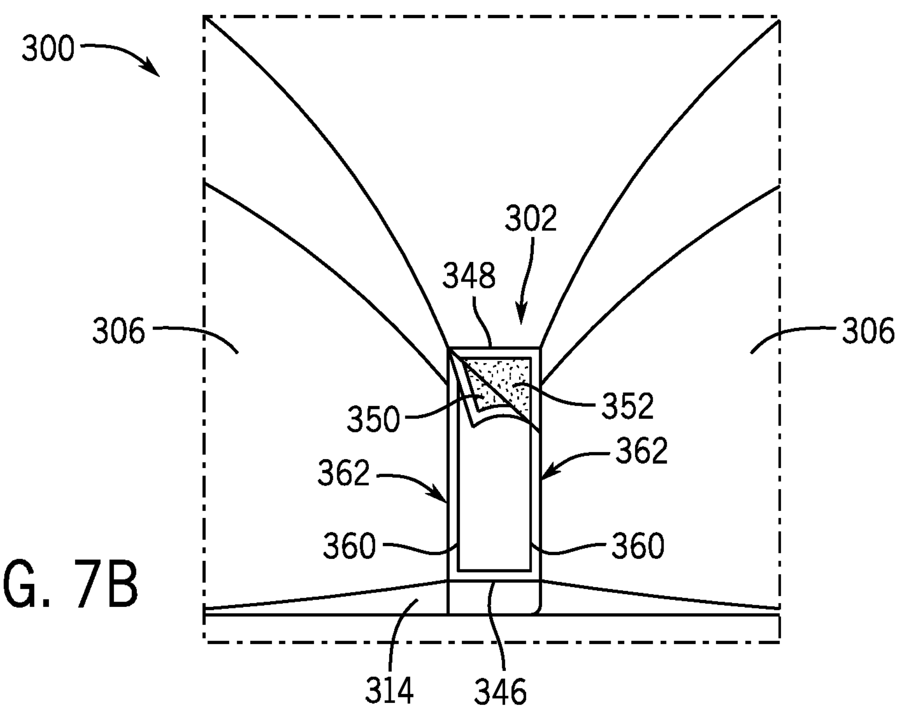
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FIG. 7B

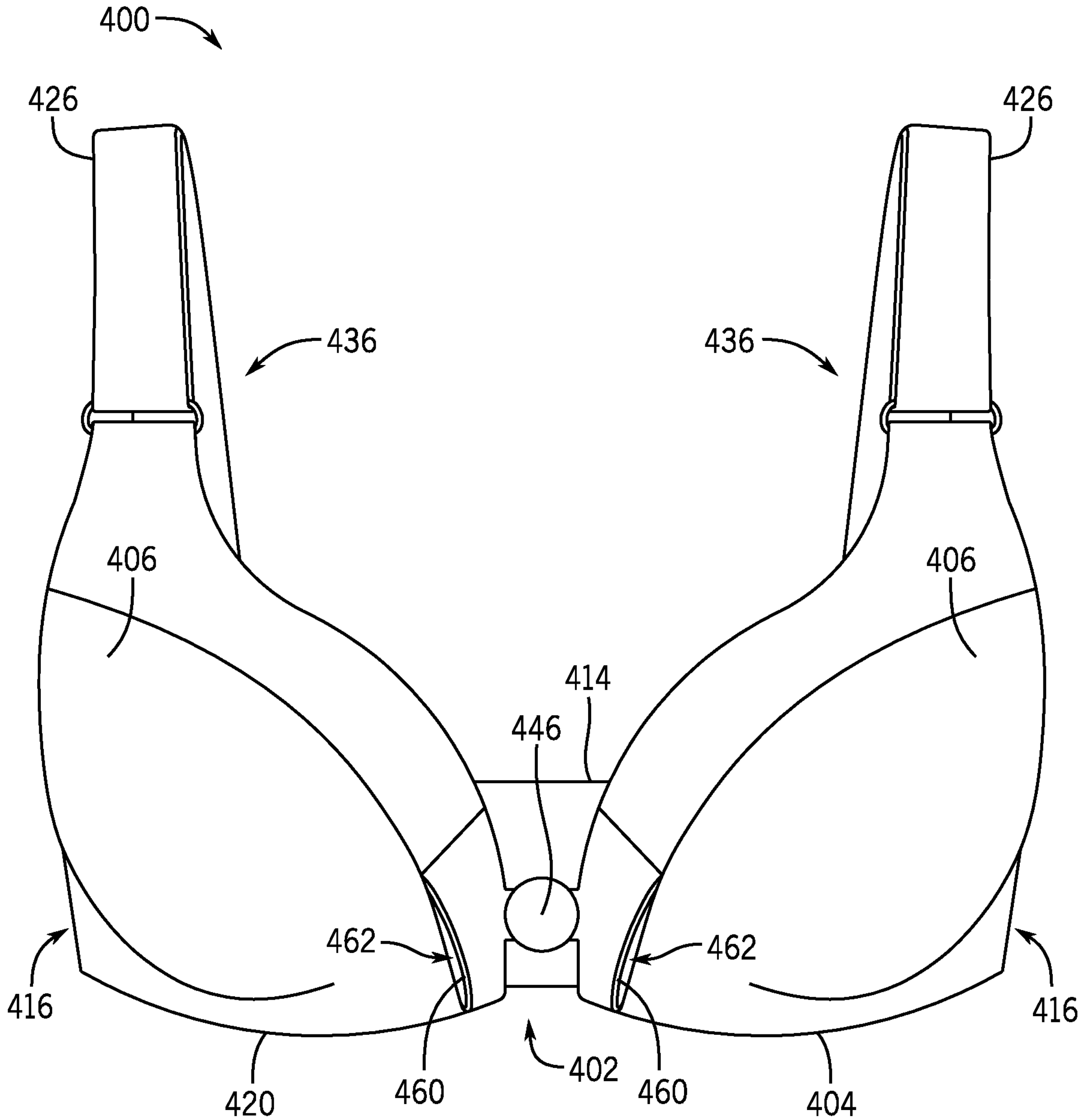
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FIG. 8

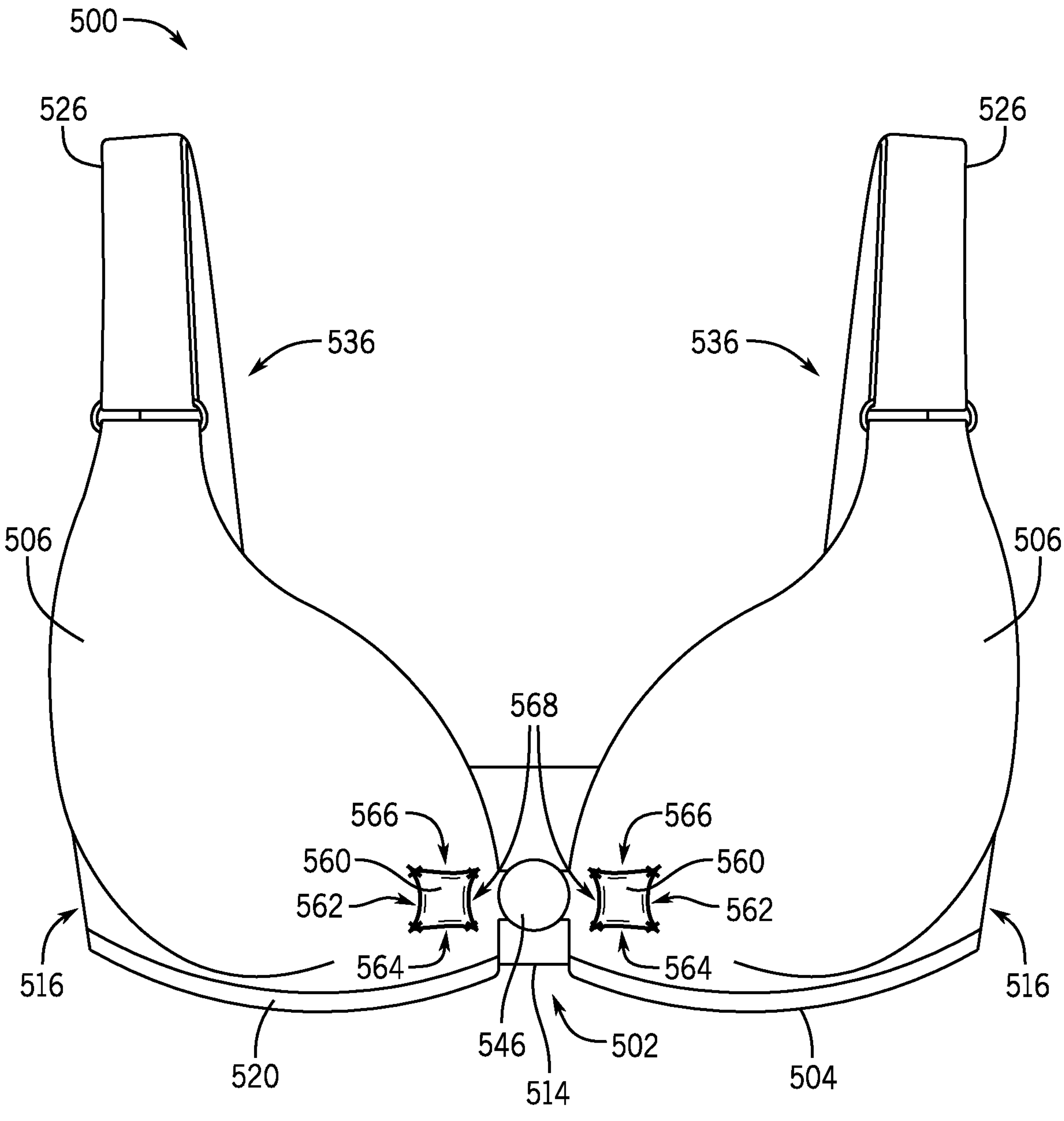
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FIG. 9

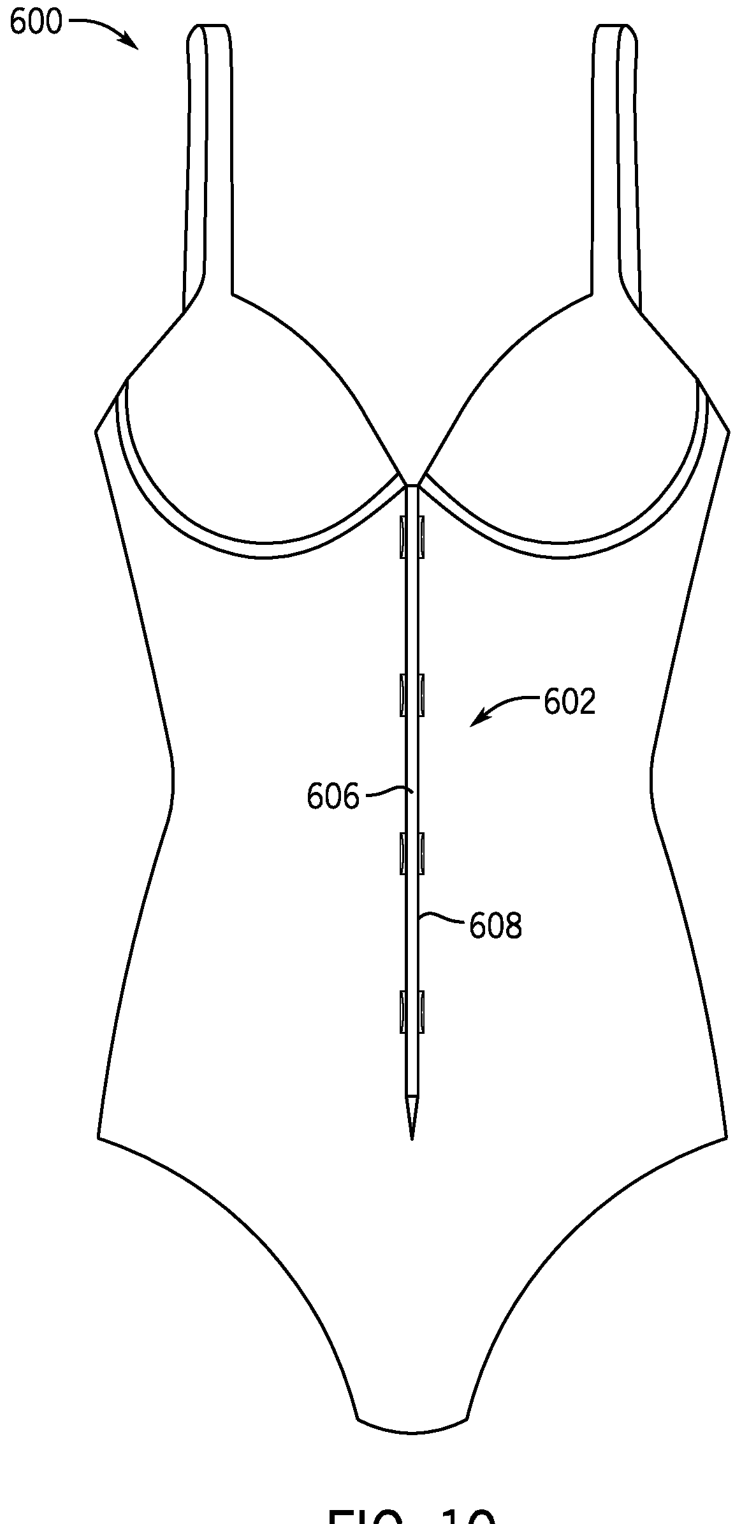

FIG. 10

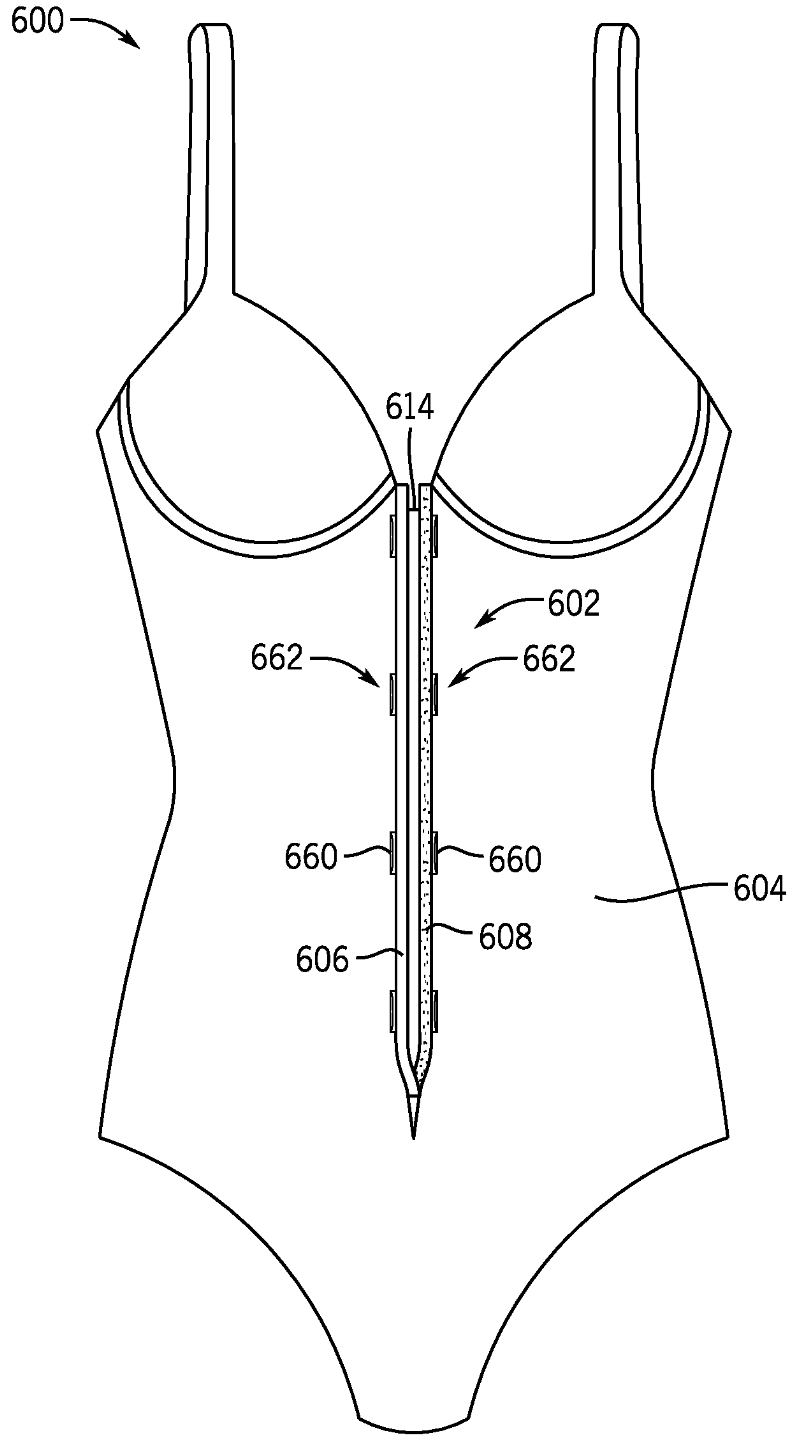

FIG. 11

SYSTEMS AND METHODS FOR A BRA FOR USE WITH LIMITED DEXTERITY

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/347,859, filed on Jun. 1, 2022, and entitled "SYSTEMS AND METHODS FOR A BRA FOR USE WITH LIMITED DEXTERITY," the entirety of which is incorporated herein by reference.

REFERENCE REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable.

SEQUENCE LISTING

Not applicable.

BACKGROUND

1. Field of the Invention

The present disclosure relates generally to an article of clothing. More particularly, the present disclosure relates to brassieres ("bras") and other undergarments and the functionality and accessibility of a closure and adjustment system thereof.

2. Description of the Background

A variety of conditions, events, or abnormalities can limit the mobility and/or dexterity of a person. In particular, certain upper body mobility impairments can include, for example, limited dexterity, limited shoulder mobility, and restricted or non-existent use of one or both arms. Limited mobility or dexterity can be caused by a variety of factors, including injury, surgery, aging, birth abnormalities, or onset of disease. As such, a variety of mobility or dexterity impairments may decrease a person's ability to easily, independently, painlessly, or successfully put on, adjust, and secure certain articles of clothing, including conventional bras. For example, conventional bras having a rear closure can require a certain level of finger dexterity to secure the closure and adjust the bra straps. Additionally, conventional bras can require cumbersome movement and rotation of shoulders to extend arms through shoulder straps of the bra. Other conventional bras can include a front closure that can similarly require certain finger dexterity and closure manipulation that can include twisting, pivoting, snapping, and/or precise clasp alignment for example, that may be difficult to secure single-handedly and/or with limited mobility or dexterity.

SUMMARY

An article of clothing, and in particular, a bra, as described herein, may have various configurations. The bra may have a closure system that includes first and second clasp members that can be coupled to one another to secure the bra. The closure system can also include one or more finger engaging members that can facilitate manipulating front panels of the bra and coupling the first and second clasp members. In some embodiments, the finger engaging members can be configured as pockets or openings that a user can insert a

finger into to urge each of the clasp members medially to secure the bra on a wearer (e.g., the user themself).

In some embodiments, the present disclosure provides a bra for securing or supporting breasts of a user with limited dexterity. The bra can include a shoulder strap adjustment system that includes first and second sliders through which each respective shoulder strap can extend through. Each shoulder strap can include a distal end that can be threaded through the respective slider and doubled back on a strap body of the respective shoulder strap. The distal end can include a first fastener half and the strap body can include a second fastener half so that the distal end can be pulled (or retracted) through the slider to shorten (or lengthen) the shoulder straps and then secured to the strap body of the shoulder strap via the fasteners. The first and second fastener halves can include a hook and loop fastener (e.g., Velcro).

In some embodiments, the present disclosure provides an undergarment configured to facilitate operation by a wearer with limited dexterity. The undergarment can include a front portion having a first panel and a second panel. The first panel can be configured to at least partially surround a first breast and the second panel can be configured to at least partially surround a second breast. The undergarment can also include a rear portion, first and second lateral side portions, a band, and a closure system. The first lateral side portion can connect the first panel of the front portion with the rear portion. The second lateral side portion can connect the second panel of the front portion with the rear portion. The band can be configured to extend circumferentially around a rib cage of the wearer. The band can be disposed along a bottom of each of the first and second panels at the front portion. The closure system can be configured to connect the band to form a closed loop. The closure system can include a first clasp member, a second clasp member, a first finger engaging member, and a second finger engaging member. The first finger engaging member can be adjacent to the first clasp member and can include an opening dimensioned to receive a first finger. The second finger engaging member can be adjacent to the second clasp member and can include an opening dimensioned to receive a second finger.

In some embodiments, the present disclosure provides a closure system for an article of clothing. The closure system can include first and second clasp members and first and second finger pockets. The second clasp member can be configured to be selectively coupled and decoupled from the first clasp member. The first and second finger pockets can each include openings and can be dimensioned to receive first and second fingers, respectively. The opening in the first finger pocket can be positioned laterally away from the first clasp member and the second finger opening can be positioned laterally away from the second clasp member so that the first and second fingers can engage the respective first and second finger pockets and move in a medial direction to move the first and second clasps toward each other to couple or decouple the closure system.

In some embodiments, the present disclosure can provide a method of securing a bra. The method can include positioning a first shoulder strap of the bra over a first shoulder of a wearer, positioning a second shoulder strap of the bra over a second shoulder of the wearer, inserting a first finger into a first finger pocket disposed on a medial portion of a first panel of the bra, the first panel configured to at least partially surround a first breast, and inserting a second finger into a second finger pocket disposed on a medial portion of a second panel of the bra, the second panel configured to at least partially surround a second breast. The method can further include moving each of the first and second fingers in a medial direction (with respect to a torso of the wearer) within the respective finger pockets to engage first and second clasp members of the bra that are disposed on the medial portions of the respective first and second panels of the bra.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a front view of a bra that facilitates operation by someone with reduced dexterity according to an embodiment of the present disclosure;

FIG. **2** is a front view of a bra that facilitates operation by someone with reduced dexterity according to another embodiment of the present disclosure;

FIG. **3** is a detailed view of the section **3**-**3** of FIG. **2** showing a finger opening;

FIG. **4** is a rear view of the bra of FIG. **2**;

FIG. **5**A is a front isometric view of a clasp assembly of the bra of FIG. **2** in an open configuration;

FIG. **5**B is a front view of the clasp assembly of FIG. **5**A in an open configuration;

FIG. **5**C is a front view of the clasp assembly of FIG. **5**A in a closed configuration;

FIG. **5**D is a front isometric view of the clasp assembly of FIG. **5**A in the closed configuration;

FIG. **6** is a front view of a bra that facilitates operation by someone with reduced dexterity according to another embodiment of the invention;

FIG. **7**A is a detailed view of a closure system of the bra of FIG. **6** in an open configuration;

FIG. **7**B is a detailed view of the closure system of FIG. **7**A in a partially closed configuration;

FIG. **8** is a front view of a bra that facilitates operation by someone with reduced dexterity according to another embodiment of the disclosure;

FIG. **9** is a front view of a bra that facilitates operation by someone with reduced dexterity according to another embodiment of the disclosure;

FIG. **10** is a front view of an undergarment that facilitates operation by someone with reduced dexterity according to another embodiment of the disclosure; and

FIG. **11** is a front view of the undergarment of FIG. **10** in an open configuration.

DETAILED DESCRIPTION OF THE DRAWINGS

The following discussion and accompanying figures disclose various embodiments or configurations of a bra and a closure system. Although embodiments of a bra or closure system are disclosed with reference to a wireless bra, concepts associated with embodiments of the bra or closure system may be applied to a wide range of undergarments and closure systems, including underwire bras, bralettes, strapless bras, nursing bras, and racerback bras, for example. Concepts of the bra or closure system may also be applied to other articles of clothing including underwear, swimwear, lingerie, and other corrective or supportive garments that may benefit from an adjustable or limited mobility closure system. Accordingly, concepts described herein may be utilized in a variety of consumer products.

The term "about," as used herein, refers to variation in the numerical quantity that may occur, for example, through typical measuring and manufacturing procedures used for articles of clothing or other articles of manufacture that may include embodiments of the disclosure herein; through inadvertent error in these procedures; through differences in the manufacture, source, or purity of the ingredients used to make the compositions or mixtures or carry out the methods; and the like. Throughout the disclosure, the terms "about" and "approximately" refer to a range of values ±5% of the numeric value that the term precedes.

As briefly described above, a variety of conditions, events, or abnormalities can limit the mobility and/or dexterity of a person. In particular, certain upper body mobility impairments can include, for example, limited dexterity or other limited mobility in digits or extremities. Limited mobility or dexterity can be caused by a variety of factors, including stroke paralysis, nerve damage, injuries, surgery, amputation, aging, birth abnormalities, arthritis, fibromyalgia, scleroderma, and other diseases. As such, a variety of dexterity impairments may decrease a person's ability to easily, independently, painlessly, or successfully put on, adjust, and secure certain articles of clothing, including bras.

For example, conventional bras having a rear closure can require a certain level of finger dexterity to secure the closure and adjust the bra straps. Additionally, conventional bras can require cumbersome movement and rotation of shoulders to extend arms through shoulder straps of the bra. Other conventional bras can include a front closure that can similarly require certain finger dexterity and closure manipulation that can include twisting, pivoting, snapping, and/or precise clasp alignment for example, that may be difficult to secure with limited mobility or dexterity. Furthermore, conventional front-closure bras can require significant grip strength to bring the two bra panels medially together in front of or beneath the breasts, which also often uses multiple fingers of each hand (e.g., both left and right thumb and index fingers) to hold the bra panels together as the user aligns the closure.

Thus, embodiments of the present disclosure address these and other drawbacks of conventional bras. For example, embodiments of the present disclosure provide a bra having a closure system configured to facilitate use by someone with reduced or limited dexterity. In particular, embodiments of the closure system described herein can provide a front closure that is accessible while wearing the bra. The closure system can provide a simplified closure mechanism that can be secured via limited digit involvement (e.g., two-finger operation such as two index fingers) or secured using a single hand (e.g., two-finger operation such as index finger and thumb of a single hand). In contrast, conventional front closures may include one or more closure systems that require a moderate degree of dexterity (e.g., more than one hand or more than two fingers, such as index finger and thumb coordination of two hands) and the application of a securing force. Such conventional front closures can include one or more of hook and eye clasps, slidable pin and groove clasps that require a certain rotation to engage and disengage the clasp ends, and snap closures that require significant force to connect and disconnect the mating halves.

Further, embodiments of the disclosure can provide a bra having adjustable straps (i.e., in a vertical direction) that can be infinitely (e.g., continuously) adjusted to tighten or loosen the shoulder straps of the bra. Conventional bra straps may be adjusted via a two-handed operation that requires pulling a portion of the strap through a metal or plastic slider. Adjustment of conventional bra straps can be time consuming and difficult to access given the slider component often is positioned at a posterior location when the user is wearing the bra. While some users may attempt to reach the difficult position of the sliders, others may attempt a trial-and-error method and make adjustments before putting on the bra, often only to find further adjustments are required.

FIG. **1** illustrates an exemplary embodiment of a bra **100** having a closure system **102** configured to be positioned at an anterior (i.e., front of the body) position of the user. The bra **100** can include a front portion **104** having first and second panels **106** that can each be generally triangularly shaped or otherwise define cup portions of the bra **100**. Typically the first panel **106** is symmetric with the second panel **106** about a central vertical axis **108**, however, other geometries are possible. In general, the cup portions of the bra can be used to surround and/or support breasts of the user. While the first and second panels **106** generally define similar geometries to that of traditional bras, other geometries and constructions are possible, including geometries that may provide more or less coverage or support to the user.

The bra **100** can further include a rear portion **114** opposite the front portion **104**. The rear portion **114** can be connected to the front portion **104** via first and second lateral side portions **116**. Each of the first and second lateral side portions **116** can at least partially define a band **120** of the bra **100**. The band **120**, similar to conventional bras, can be disposed along a bottom of each of the first and second panels **106**. In general, the band **120** is configured to surround (e.g., extend circumferentially around) the rib cage of the wearer below the breasts and can define at least one dimension (e.g., band size) used by a wearer to determine the fit of the bra **100**. In a secured configuration of the bra **100**, the band **120** forms a continuous circuit or closed loop around the torso of the user and extends generally horizontally below their breasts. However, it should be appreciated that different torso geometries can affect the exact orientation or contours of the band **120**. Additionally, in some embodiments, the band **120** or portions of the band **120** can include an elastic material to further provide adjustment and comfort of the bra **100**.

The bra **100** can further include first and second shoulder straps **126**. Each of the first and second shoulder straps **126** also connect the front portion **104** and the rear portion **114**. In particular, each of the first and second shoulder straps **126** can extend from a top portion of the respective first and second panels **106**. In some embodiments, the first and second panels **106** can be integrally formed with or otherwise affixed to the rear portion **114**. Additionally or alternatively, the first and second shoulder straps **126** can be integrally formed or otherwise fastened to the respective first and second panels **106**.

In some embodiments, the shoulder straps **126** may be adjustable via an adjustment system **136**. The adjustment system **136** can be configured to adjust the shoulder straps **126** independently of one another in a vertical direction to lengthen or shorten the shoulder straps **126** of the bra **100**. In the illustrated embodiment, the adjustment system **136** can include first and second sliders **138** at the respective first and second shoulder straps **126**. The sliders **138** may act as a link between the first and second panels **106** and the respective first and second shoulder straps **126**.

In use, the first shoulder strap **126** may be threaded through the first slider **138** and folded back on itself so that the first shoulder strap **126** forms a loop adjacent to the top portion of the first panel **106** and a distal end of a strap body of the shoulder strap **126** extends toward the rear portion **114** of the bra **100**. In some embodiments, the distal end of the shoulder strap **126** may terminate proximate to the top of the wearer's shoulder. The user can then separate (e.g., pull away) the distal end of the shoulder strap **126** from a proximal portion of the shoulder strap **126** and pull or retract the shoulder strap **126** through the first slider **138** to tighten or loosen the shoulder strap **126**. In some embodiments, the distal end of the shoulder strap **126** may be selectively secured to a proximal portion of the shoulder strap **126** via a hook and loop fastener (e.g., Velcro®). The second shoulder strap **126** may be similarly adjusted in length via the second slider **138**.

With continued reference to FIG. **1**, the closure system **102** can include a first clasp member **146** and a second clasp member **148**. The first clasp member **146** can be secured to the first panel **106** at a medial portion near the band **120** and the second clasp member **148** can be secured to the second panel **106** at a medial portion near the band **120**. In particular, the first clasp member **146** can extend in a medial direction **150** (i.e., toward a center of the user, bra **100**, and axis **108**) from the first panel **106** and the second clasp member **148** can extend in the medial direction **150** from the other panel **106**. In use, the first clasp member **146** can be inserted into the second clasp member **148** (or vice versa) to secure the bra **100** and form a continuous circuit at the band **120**. Advantageously, the closure system **102** can be secured without requiring any twisting, turning, or pivoting of the medial sections (i.e., adjacent to the axis **108**) of the first and second panels **106**. Furthermore, the closure system **102** can be secured using a single hand. For example, a single hand can pinch finger engaging members **160**, which can be configured as pockets or other members, to couple (or decouple) the clasp members **146**, **148**.

In general, fastening the closure system **102** of the bra **100** promotes quick and simple bra securement, especially for users with reduced dexterity, such as limited or non-existent finger, wrist, or arm movement, for example. Similarly, releasing the closure system **102** of the bra **100** can be accomplished by a user with reduced dexterity. In the illustrated embodiment, a portion of the first clasp member **146** may be pressed inward (e.g., in a posterior direction) or otherwise disengaged from the second clasp member **148** without requiring the user to bend, twist, or rotate any part of the closure system **102** or first and second panels **106**. For example, a force exerted in a single direction (e.g., posterior) can be used to release the closure system **102** and then little or no force may be required to separate the first and second clasp members **146**, **148** in a second direction that is generally perpendicular to the first direction. The second direction can be a lateral direction **152**. When the closure system **102** is in a released orientation, the bra **100** can be removed from (or put on) the user in a similar motion to removing (or putting on) a vest.

In some embodiments, the closure system **102** can include a magnet. The magnet can be used to facilitate a preliminary alignment of the first and second clasp member **146**, **148** or can be incorporated into the first and second clasp members **146**, **148** so that the closure system **102** remains in a closed position via a magnetic force. In general, the magnet within the closure system **102** can advantageously help align the medial sections of the first and second panels **106**, especially one-handedly, so that a user can effectively and efficiently secure the bra **100**. In some embodiments, the magnet strength of the magnet within the closure system **102** can be tailored to an individual's preference or need. By way of example, a user with limited dexterity and only a single hand or arm may benefit from a different-strength magnetic closure system **102** than another user with carpal tunnel syndrome.

FIG. **2** illustrates another embodiment of a bra **200** having a closure system **202**. Like the bra **100**, the closure system

202 of the bra 200 is configured to be positioned at an anterior position of the user when the user is wearing the bra 200. The bra 200 includes a front portion 204 having first and second panels 206 that can be generally triangularly shaped and can define cup portions of the bra 200. The bra 200 can further include a rear portion 214 opposite the front portion 204. The rear portion 214 can be connected to the front portion 204 via first and second lateral side portions 216. The first and second lateral side portions 216 can at least partially define a band 220 of the bra 200.

The bra 200 can further include first and second shoulder straps 226 that connect the front portion 204 and the rear portion 214. The shoulder straps 226 can each define respective distal ends 228 and a strap body 230. In the illustrated embodiment, the strap body 230 of the shoulder straps 226 can extend between the distal end 228 and the rear portion 214 of the bra 200. Additionally, the shoulder straps 226 can include a fold or a loop 232 in the strap body 230 adjacent to the top portion of the respective first and second panels 206.

In some embodiments, the shoulder straps 226 may be adjustable via an adjustment system 236. The adjustment system 236 can be configured to adjust the shoulder straps 226 independently of one another in a vertical direction to lengthen or shorten the shoulder straps 226 of the bra 200. In the illustrated embodiment, the adjustment system 236 can include first and second sliders 238 at the loop 232 of the shoulder straps 226. The sliders 238 may act as a link between the first and second panels 206 and the respective first and second shoulder straps 226.

In use, one of the shoulder straps 226 may be threaded through the slider 238 and folded back on itself so that the distal end 228 extends toward the rear portion 214 of the bra 200. In some embodiments, the distal end 228 of the shoulder strap 226 may terminate proximate to the top of the wearer's shoulder, or otherwise at an anterior position of the wearer so that the user can access the distal end 228 without reaching behind their shoulder. To lengthen or shorten the shoulder straps 226, the user can grab the distal end 228 of the shoulder strap 226 and separate (e.g., pull away) the distal end 228 from the strap body 230. The user can then pull or retract the shoulder strap 226 through the slider 238 to tighten or loosen the strap 226.

In some embodiments, one side of the strap body 230 of the shoulder strap 226 can include a first fastener portion and one side of the distal end 228 of the shoulder strap 226 can include a second fastener portion configured to engage and be secured with the first fastener portion. For example, one side 240 of the strap body 230 can include one of a hook and loop material of a hook and loop fastener (e.g., Velcro), and one side of the distal end 228 can include the other of the hook and loop material of the hook and loop fastener. In other embodiments, the adjustment system 236 can include a magnetic fastener. Thus, in use, the user can selectively couple the distal end 228 to the strap body 230 of the respective shoulder strap 226 to set their preferred strap length, particularly while wearing the bra 200.

In the illustrated embodiment, the distal end 228 further includes a grip portion 242. The grip portion 242 can be configured as a loop, pocket, opening, or protrusion in the distal end 228 to facilitate gripping or manipulating the distal end 228 to separate the distal end 228 from the strap body 230 to adjust the lengths of the shoulder straps 226. An opening provided by the grip portion 242 can help a user with limited dexterity to separate the distal end 228 from the strap body 230. For example, a user with limited finger dexterity may have trouble closing their fingers together such that grasping items is difficult. The grip portion 242 can allow the user to slide one or more fingers through the distal end 228 and allow them to pull the distal end 228 away from the strap body 230 without requiring a grasping (e.g., closing one's fingers) motion.

With continued reference to FIG. 2, the closure system 202 can include a first clasp member 246 and a second clasp member 248. The first clasp member 246 can be secured to the first panel 206 at a medial portion near or in line with the band 220 and the second clasp member 248 can be secured to the second panel 206 at a medial portion near or in line with the band 220. In use, the first clasp member 246 can be inserted into, or otherwise secured relative to, the second clasp member 248 to secure the bra 200 and form a continuous loop at the band 220. Advantageously, the closure system 202 can be secured generally without requiring twisting, turning, or pivoting of the first and second clasp members 246, 248 nor the first and second panels 206.

The closure system 202 can further include finger engaging members 260 configured as finger pockets having one or more finger openings. The finger engaging members 260 can be disposed on the medial portions of the panels 206 adjacent to respective first and second clasp members 246, 248. In particular, the finger engaging members 260 can be formed along or defined by the band 220. In use, a user can insert one finger into each finger engaging member 260 to manipulate the medial portions of the panels 206 and move the first clasp member 246 toward the second clasp member 248. For example, the user can insert a first finger (e.g., a right index or middle finger) into a first finger engaging member 260 and a second finger (e.g., a left index or middle finger) into a second finger engaging member 260 and urge each of the panels 206 medially (with respect to their body) so that the first clasp member 246 is urged toward and eventually engaged with and secured to the second clasp member 248. In another example, a user can use a single hand to engage each of the finger engaging members 260 (e.g., an index finger and a thumb of the same hand). However, it should be appreciated that other combinations of digits or appendages can be used to operate the closure system 202.

Furthermore, similar to the closure system 102, the closure system 202 can include one or more magnets. The magnets can be used to facilitate a preliminary alignment or engagement of the clasp members 246, 248. For example, the magnets can help attract and align the clasp members 246, 248 before the clasp members are in a locking engagement. Once the clasp members 246, 248 are aligned, they can be moved into a sliding engagement with each other to close and secure the bra 200. In the illustrated embodiment, the clasp members 246, 248 are disposed medially inward from the finger engaging members 260. That is, the finger engaging members 260 are positioned laterally outward from the clasp members 246, 248 so that when the closure system 202 is in a closed position, as shown in FIG. 2, the finger engaging members 260 and the clasp members 246, 248 extend in a transverse direction across the front of the bra 200 and are not stacked on top of each other. Thus, in a closed position, the finger engaging members 260 and the clasp members 246, 248 may be positioned along a circuit generally defined by the bra 200 band 220.

With reference to FIG. 3, the finger engaging member 260 can provide one or more openings. Though only one of the two finger engaging members 260 of the bra 200 is shown, it should be appreciated that the other finger engaging member 260 can have similar (e.g., symmetric) openings. In the illustrated embodiment, the finger engaging member 260 provides a first opening **262** and a second opening **264**. The arrows in FIG. **3** indicate the direction in which a finger may be inserted into the finger engaging member **260** to move the first and second clasp members **246**, **248** toward each other.

For example, a finger may be inserted into the first opening **262** and the closing motion for closing the bra **200** may include pointing the first finger from a right hand toward a second finger from a left hand (similarly inserted into the other finger engaging member **260**) and moving the right and left hands medially under (or across, or near) the breasts of the user. Alternatively, a finger may be inserted into the second opening **264** and the closing motion for closing the bra **200** may include pointing the first finger from a right hand upward and pointing a second from a left hand (similarly inserted into the other finger engaging member **260**) upward and moving the right and left hands medially so that the first and second fingers are generally parallel. In other examples, a single hand can be used so that two fingers of one hand can engage either of the openings **262**, **264** on each panel **206** and move in a pinching motion to close the bra **200**.

In general, each finger engaging member **260** promotes single finger operation and does not require the user to grasp a single element of the closure system **202** between two fingers of one hand, which is often required on closure systems (front or rear) of conventional bras. Additionally, while only two openings **262**, **264** are shown in the illustrated embodiment, additional (or fewer) openings are possible (see, for example, FIG. **9**). Furthermore, as shown in FIG. **3**, the finger engaging member **260** can be configured as a piece of material secured at the band **220** of the bra **200**. However, in other embodiments, the finger engaging member **260** can be more integral with the band. For example, the band **220** may be generally sewn as a tube, and a medial end of the tube may be opened and folded laterally (e.g., away from a central axis, opposite the medial direction) and sewn or otherwise affixed on the sides to form a pocket.

FIG. **4** illustrates a rear review of the bra **200**. In some embodiments, the rear portion **214** can optionally include a rear closure system **270**. In some embodiments, the rear closure system **270** may be in addition to or in lieu of the front closure system **202**. Alternatively, the front closure system **202** may be the only closure system of the bra **200** and the rear portion **214** is permanently fixed (or integrally formed) so that the bra **200** is not separatable at the rear portion **214**. Further, in some embodiments, the rear closure system **270** can also include one or more finger engaging members **260***a* (though only one finger engaging member **260***a* is visible in FIG. **4**). Similar to the finger engaging member **260**, the rear finger engaging members **260***a* can be a pocket sewn or otherwise formed on the band **220** that can be engaged by a digit to facilitate a closing operation on either side of the rear closure system **270**.

FIGS. **5**A-D illustrate the first and second clasp members **246**, **248** of the closure system **202** according to an exemplary embodiment. As shown, the first and second clasp members **246**, **248** can form a flower design that is separable when the closure system **202** is in an open or disengaged orientation; however, alternative designs are contemplated. In general, fastening the closure system **202** of the bra **200** facilitates quick and easy bra securement, especially for users with reduced dexterity, such as limited finger, wrist, or arm movement. Similarly, releasing the closure system **202** of the bra **200** can be accomplished by a user with reduced dexterity.

In the illustrated embodiment, the first clasp member **246** can include a first lock body or lock member **274** and the second clasp member **248** can include a corresponding second lock body or lock member, such as an opening **276**, for example. The lock member **274** of the first clasp member **246** may be pressed inward or otherwise disengaged from an opening **276** of the second clasp member **248** without requiring the user to bend, twist, or rotate any part of the closure system **202** or the first and second panels **206** of the bra **200**. For example, a force exerted in a single first direction on the lock member **274** can be used to release the closure system **202** and then little or no force may be used to separate the first and second clasp members **246**, **248** in a direction that is generally perpendicular to the first direction. In general, FIGS. **5**A and **5**B illustrate the first and second clasp members **246**, **248** in an open, unsecured configuration and FIGS. **5**C and **5**D illustrate the first and second clasp members **246**, **248** in a closed, secured configuration.

In general, the first and second lock members can provide an engagement (e.g., a sliding engagement) between the first and second clasp members **246**, **248**. Thus, the first and second clasp members **246**, **248** can be selectively coupled and decoupled from each other. As described above, the first and second clasp members **246**, **248** can be coupled by being slid or otherwise pinched together in the medial direction.

Furthermore, in other embodiments, the closure system **202** can be released by pinching the first and second clasp member **246**, **248** inward (e.g., medially) to disengage the lock member **274**. Thus, the closure system **202** can be coupled and decoupled with the same or similar movement. Additionally, as described above, the closure system **202** can include magnets to facilitate securing and closing the bra **200**. For example, the clasp members **246**, **248** can include magnets therein that help align the clasp members **246**, **248** to engage (e.g., slidingly engage) one another and move into a locked position so that the bra **200** is clasped, closed, and secured. In general, a magnetic closure can be used to reduce the amount of precision required to close the bra **200** by self-centering or aligning the clasp members **246**, **248**, which can be beneficial to those with limited dexterity or singular limbs.

FIG. **6** illustrates another embodiment of a bra **300** having a closure system **302**. The bra **300** includes similar components and features as the bras described herein, thus, descriptions of these components, identified with like reference numbers, can be found above with reference to the bras **100** and **200**. Such similar components and features of the bra **300** include a front portion **304**, first and second panels **306**, a rear portion **314**, first and second lateral side portions **316**, a band **320**, and first and second shoulder straps **326**. The bra **300** can optionally include an adjustment system **336** at the shoulder straps **326** to adjust the length of the shoulder straps **326**.

As shown in FIGS. **6**, **7**A, and **7**B, the closure system **302** of the bra **300** can include a first clasp member **346** and a second clasp member **348**. The first and second clasp member **346**, **348** can be configured as a variety of closure systems, including, for example, a hook and loop closure, a mechanical clasp having a sliding engagement or interference fit, and/or a magnet. Each clasp member **346**, **348** can include an engagement portion that interacts with the corresponding engagement portion of the other clasp member. For example, as shown in FIG. **7**B, the first clasp member **346** can include an engagement portion **350** that faces toward the wearer and the second clasp member **348** can include a corresponding engagement portion **352** that faces away from the wearer (e.g., when the bra **300** is in a closed configuration).

In one embodiment, the first engagement portion **350** can be configured as a loop material and the second engagement portion can be configured as a hook, material (or vice versa) of a hook and loop closure. As indicated in FIG. 7B, when the closure system **302** is in a closed position, a lower portion of the bra **300** (i.e., the band **320** or adjacent to the band **320**) forms a complete circuit. In this regard, the bra **300** looks like a conventional rear-closure bra when in a closed position such that the first and second clasp member **346**, **348** are not visible (e.g., in contrast to the closure system **202** described above).

Additionally, the closure system **302** can include a pair of finger engaging members **360** configured as finger pockets having one or more finger openings **362**. The finger engaging members **360** can be disposed on the medial portions of the panels **306** adjacent to (e.g., on top of or under) the respective first and second clasp members **346**, **348** and the band **320**. As shown in FIG. 7B, in the illustrated embodiment, the finger engaging members **360** generally coincide with a respective one of the clasp members **346**, **348**. That is, when the closure system **302** is in a closed position, the finger engaging members **360** and the clasp members **346**, **348** are stacked and are positioned at the same (or close) position along a medial/distal axis that extends along a transverse plane of the user. Likewise, when the closure system **302** is in a closed position, the finger engaging members **360** and the clasp members **346**, **348** each are positioned along, or at least partially overlap a central vertical axis (e.g., like axis **108** shown in FIG. **1**).

Furthermore, in the illustrated embodiment, the finger engaging members **360** define an outer periphery **364** (e.g., a closed, rectangular periphery, however, other geometries are possible including triangles or semi-circles). As shown, the outer periphery **364** defines a gap between the outer periphery **364** and the edge of the bra **300** at the general medial area of the panels **306**. The outer periphery **364** can define the pocket formed by the finger engaging members **360**. Furthermore, the outer periphery **364** can include multiple sides. For example, in the illustrated embodiment, the outer periphery **364** defines four sides. One of the sides of the outer periphery (e.g., a first side) can define the opening **362** and the remaining sides can be closed to form a pocket. In other embodiments (e.g., see FIG. **9**), a finger engaging member, such as the finger engaging member **560** can still define an outer periphery and also include a plurality of openings.

In use, the user can insert a first finger **370** into the opening **362** at the first clasp member **346** and a insert a second finger **372** into the opening **362** at the second clasp member **348** to manipulate the medial portions of the panels **306** and move the first clasp member **346** toward the second clasp member **348**. The closing motion for closing the bra **300** can include pointing the first finger **370** toward the second finger **372** and moving the fingers (e.g., hands) medially under (or across, or near) the user's breasts. The fingers **370**, **372** can generally move in parallel so that the loop portion **350** (or other closure type) of the first clasp member **346** can engage the hook portion **352** (or other corresponding closure type) of the second clasp member **348** to secure the closure system **302** of the bra **300**, as shown generally in FIG. 7B. In other exemplary closing motions, two fingers from a single hand (e.g., an index finger and thumb) can be used to close the closure system **302** by moving the fingers in a closing or pinching motion. Additionally, the finger openings **362** can be used to disengage the first and second clasp members **346**, **348**.

FIG. **8** illustrates another example embodiment of a bra **400** having a closure system **402**. The bra **400** includes similar components and features as the bras described herein, thus, descriptions of these components, identified with like reference numbers, can be found above with reference to the bras **100** and **200**. Such similar components and features of the bra **400** include a front portion **404**, first and second panels **406**, a rear portion **414**, first and second lateral side portions **416**, a band **420**, and first and second shoulder straps **426**. The bra **400** can optionally include an adjustment system **436** at the shoulder straps **426** to adjust the length of the shoulder straps **426**.

The closure system **402** of the bra **400** can include first and second clasp members **446**, which may include one or more of a hook and loop closure, a button closure, or other mechanically or magnetically engaging closures. Additionally, the closure system **402** can include a pair of finger engaging members **460** configured as finger pockets having one or more finger openings **462**. The finger engaging members **460** can be disposed on the medial portions of the panels **406** adjacent to the respective first and second clasp members **446** and the band **420**. In the illustrated embodiment, the finger engaging members **460** are positioned laterally from the clasp members **446**. The finger openings **462** can be used similarly as described above with reference to the bras **200** and **300** to secure the first and second clasp members.

FIG. **9** illustrates another example embodiment of a bra **500** having a closure system **502**. The bra **500** includes similar components and features as the bras described herein, thus, descriptions of these components, identified with like reference numbers, can be found above with reference to the bras **100** and **200**. Such similar components and features of the bra **500** include a front portion **504**, first and second panels **506**, a rear portion **514**, first and second lateral side portions **516**, a band **520**, and first and second shoulder straps **526**. The bra **500** can optionally include an adjustment system **536** at the shoulder straps **526** to adjust the length of the shoulder straps **526**. In some embodiments, the adjustment system **536** can include a finger loop that can allow a user to insert a digit therethrough to adjust the length of each shoulder strap **526**.

The closure system **502** of the bra **500** can include first and second clasp members **546**, which may include one or more of a hook and loop closure, a button closure, or other mechanically or magnetically engaging closures. Additionally, the closure system **502** can include a pair of finger engaging members **560** configured as finger pockets having one or more finger openings **562**, **564**, **566**, **568**. The finger engaging members **560** can be disposed on the medial portions of the panels **506** adjacent to the respective first and second clasp members **546** and the band **520**. The finger openings **562**, **564**, **566**, **568** can be used similarly as described above with reference to the bra **200** to secure the first and second clasp members. Additionally, the finger openings **568** can be used to disengage the first and second clasp members **546**. In some embodiments, a variety of combinations of the finger openings can be used to disengage the clasp member **546**, by, for example, moving (e.g., pinching) the finger engaging members **560** medially. Thus, the disengaging motion may be the same direction as the engaging motion so that, advantageously, the same (or similar) motion can be used to open and close the bra **500**.

FIGS. **10** and **11** illustrate another example embodiment of a garment **600**. In some embodiments, the garment **600** may be worn as, for example, swimwear, lingerie, or a leotard. The garment **600** can include similar components and features as the bras described herein. In particular, the garment **600** can include a closure system **602** configured to secure a first portion **606** to a second portion **608** of the garment **600**. Notably, the closure system **602** can include a finger engaging member **660** having one or more finger openings **662**. The finger engaging members **660** can be disposed on medial portions of the first and second portions **606**, **608**. The finger openings **662** can be used similarly as described above to secure the first and second portions **606**, **608** in a closed configuration. In the illustrated example, the closure system **602** is disposed on a front portion **604** of the garment **600**. However, in other embodiments, the closures system **602** can be disposed on a rear portion **614** of the garment **600**.

In other garments, such as, for example, underwear, tights, leggings, or other clothing having waistbands or general closures, finger engaging members configured as pockets can be used to facilitate closure with limited dexterity or other limb restrictions (e.g., a user having only one (usable) arm or hand). The finger pockets can provide a grip portion so that only two fingers or a single hand are required to close the garment. Similar to the embodiments described above, closures according to embodiments of the present disclosure can be operated with minimal dexterity requirements, such as, for example, a simple pinching motion.

The present disclosure is directed to an article of clothing, and/or specific components and closure systems for an article of clothing, such as a bra and other garments and their respective closure systems. The bra or other garments may comprise one or more fabric materials, such as, for example, nylon, polyester, cotton, Spandex (e.g., elastane), silk, elastic, lyocell fibers, modal fibers, or other materials and/or fibers generally made by knitting, weaving, or forming. The closure systems described herein may comprise a variety of materials including, for example, metals and polymers (e.g., plastic), which can have varying properties (e.g., flexible, magnet, rigid, pliable, etc.) or varying visual characteristics.

Any of the embodiments described herein may be modified to include any of the structures, elements, or methodologies disclosed in connection with different embodiments. In this regard, for example, the clasps **246**, **248** of FIG. **2** (or others) may be incorporated into the closure system **302** of FIG. **6**. Additionally, it should be appreciated that a clasp, as described herein, can include a variety of systems for fastening things together. For example, a clasp can include a device with interlocking parts (e.g., sliding engagement, snap, or interference fits), magnets, hook and loop, and combinations thereof. Further, the present disclosure is not limited to bras of the type specifically shown. Still further, aspects of the bra and methods of using the bra according to any of the embodiments disclosed herein may be modified to work with any type of article of clothing or supportive or form-fitting undergarment.

As noted previously, it will be appreciated by those skilled in the art that while the invention has been described above in connection with particular embodiments and examples, the invention is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications and departures from the embodiments, examples and uses are intended to be encompassed by the claims attached hereto. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein. Various features and advantages of the invention are set forth in the following claims.

INDUSTRIAL APPLICABILITY

The systems of a methods of a bra and closure system as described herein advantageously provide an undergarment, and more specifically, a supportive undergarment, having enhanced adjustment and securement capabilities that can be utilized by an end user having limited dexterity.

Numerous modifications to the present invention will be apparent to those skilled in the art in view of the foregoing description. Accordingly, this description is to be construed as illustrative only and is presented for the purpose of enabling those skilled in the art to make and use the invention and to teach the best mode of carrying out same. The exclusive rights to all modifications which come within the scope of the appended claims are reserved.

The invention claimed is:

1. An undergarment configured to facilitate operation by a wearer with limited dexterity, the undergarment comprising:

a front portion having a first panel and a second panel, the first panel being configured to at least partially surround a first breast and arranged on a first side of a vertical axis, the second panel being configured to at least partially surround a second breast and arranged on a second side of the vertical axis;

a rear portion;

a first lateral side portion connecting the first panel of the front portion and the rear portion;

a second lateral side portion connecting the second panel of the front portion and the rear portion;

a band configured to extend circumferentially around a rib cage of the wearer, the band disposed along a bottom of each of the first and second panels at the front portion; and

a closure system configured to connect the band to form a closed loop in a closed configuration, the closure system including:

a first clasp member provided in line with the band along the bottom of the first panel and defining a first lock body;

a second clasp member provided in line with the band along the bottom of the second panel and defining a second lock body that corresponds to the first lock body;

a first finger engaging member arranged laterally outward from or on top of the first clasp member and having an opening dimensioned to receive a first finger; and

a second finger engaging member arranged laterally outward from or on top of the second clasp member and having an opening dimensioned to receive a second finger,

wherein, in the closed configuration, the first finger engaging member overlaps with the first clasp member and the second clasp member along a horizontal axis that extends between a left side of the front portion and a right side of the front portion.

2. The undergarment of claim **1**, wherein the first clasp member extends toward the vertical axis from a right portion of the first panel and the second clasp member extends toward the vertical axis from a left portion of the second panel, and

wherein each of the first and second finger engaging members are formed along the band.

3. The undergarment of claim **1**, wherein, in the closed configuration, the first clasp member is positioned on top of the second clasp member.

4. The undergarment of claim **1**, wherein the first clasp member is configured to slidably engage the second clasp member.

5. The undergarment of claim 4, wherein the closure system includes a magnet configured to facilitate alignment of the first and second clasp members prior to engagement.

6. The undergarment of claim 1, wherein an engagement portion of the first clasp member is configured to face toward the wearer and an engagement portion of the second clasp member is configured to face away from the wearer.

7. The undergarment of claim 1, wherein the opening of the first finger engaging member and the opening of the second finger engaging member are each defined by a respective outer periphery having first, second, and third sides, and

wherein the first side defines the opening in the respective finger engaging member and the second and third sides are closed.

8. The undergarment of claim 7, wherein the respective outer periphery further includes a fourth side that is closed.

9. The undergarment of claim 1, wherein each of the first and second finger engaging members further comprises at least one additional opening, such that each of the first and second finger engaging members have a plurality of openings.

10. The undergarment of claim 1, wherein the undergarment further comprises:

first and second shoulder straps, each of the shoulder straps defining a strap body and a distal end, the distal end extending toward the rear portion of the undergarment and configured to be pulled away from the strap body to lengthen or tighten the respective shoulder strap,

wherein the first shoulder strap connects the first panel to the rear portion and the second shoulder strap connects the second panel to the rear portion.

11. The undergarment of claim 1, wherein the front portion, the rear portion, and the first and second lateral side portions comprise fabric selected from the group consisting of cotton, nylon, polyester, elastane, and silk, and

wherein the first and second clasp members comprise plastic or metal.

12. The undergarment of claim 1, wherein the undergarment is a form-fitting undergarment configured as a brassiere, a swimwear, a lingerie, or a leotard.

13. The undergarment of claim 1, wherein the first clasp member and the second clasp member each include a magnet.

14. The undergarment of claim 1, wherein the first clasp member is configured to magnetically engage the second clasp member.

15. The undergarment of claim 1, wherein the first finger engaging member and the second finger engaging member are each at least partially defined by the band.

16. The undergarment of claim 1, wherein the first finger engaging member and the second finger engaging member are attached to the band.

17. The undergarment of claim 1, wherein the first clasp member and the second clasp member each include an interlocking part that is configured to provide a snap fit.

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